Verified adjoint gradients reach validated torque-ripple targets sooner in a nonlinear motor model

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Abstract

Verified design gradients matter only if they shorten the path to a design that survives independent validation. We reconstruct a published nonlinear permanent-magnet motor in our GEODE finite-element implementation, verify its discrete design gradients, and test them in a preregistered optimization comparison. In 100 budget-matched runs judged by a common order-93 validation, adjoint SLSQP reached torque-ripple targets sooner than COBYQA at the same training resolution in all 45 paired comparisons where both succeeded, by geometric means of 2.45–7.79. A coarser training mesh saved only 1.3–2.2 times and, with SLSQP, missed targets from one start in each design space. Every selected target crossing still met its target on a finer mesh, which changed the objective by -4.0% to $+1.1\%$. The objective itself depends on the numerical interface. At the finest mesh, truncating harmonic rotor–stator coupling at the reference's order 45 lowers loaded torque standard deviation by 7–9% and raises cogging standard deviation by 59% relative to conforming coupling. Retaining every order below the trace Nyquist limit reproduces conforming torque within a few $10^{-9}\,\mathrm{N\,m}$ on the two checked meshes. Refining the independent reference reduces its cogging discrepancy with GEODE from 12.9% to 0.10%, although neither spatial-refinement sequence is fully converged. For the saved optimized design, raising the interface order from 45 to 93 multiplies the squared-ripple objective by 3.6 under fixed unscaled current, while its reduction stays above 2,400-fold; at matched reference current its full-current mean agrees within 0.1%, but its ripple is 2.0 times the saved value. Of 704 distinct loaded derivative comparisons, 688 agree with best-step central differences within 2.8×10^{-7} relatively. Ripple sensitivities nevertheless differ between verified model–estimator pairs, and fixing winding-group-1 density instead of current reverses three mean-torque sensitivities at the initial design on the second-coarsest mesh, at order 45 and full current.

1 Introduction

A small error in mean torque can conceal a large change in the quantity being optimized. Reconstructing the initial motor of Wiesheu et al. (2024), we find that truncating the interface at order 45 lowers loaded torque standard deviation by 7–9% and raises cogging standard deviation by 59% on the finest mesh, relative to conforming coupling. The loaded mean torque of either model agrees with the saved reference within 0.16%. Varying the transmitted order and checking the full-order identity distinguish this model dependence from spatial refinement error. Rerunning and refining the independent reference accounts for the initial cogging discrepancy; checking the saved optimized geometry shows that interface order also affects the reported design improvement.

The same distinction applies to derivatives. An adjoint can produce many design sensitivities without a field solve per parameter (Scotzniovsky et al., 2025), but it differentiates the chosen discrete model and excitation policy. It does not establish that the objective is resolved, or that following its gradients beats a cheaper design model. These questions require separate checks of the field model, derivatives and optimization outcome.

We report a four-mesh comparison of conforming and truncated harmonic coupling, including two torque estimators and a harmonic-order ladder. We also verify derivatives of mean torque and ripple with respect to 12 design parameters at initial and perturbed designs, and expose the effect of holding current rather than current density fixed. The comparison shows how interface truncation and excitation policy affect the objective and its sensitivities even when the discrete derivatives pass verification. We then compare five optimizers in a preregistered, budget-matched study at a common validation model, separating optimizer choice from training resolution, and check the selected designs on finer meshes and angle grids. Wherever both reach a target from the same start, adjoint SLSQP gets there 2.45–7.79 times sooner, by geometric mean, than COBYQA trained on the same mesh. The numerical methods are established; the contribution is the measured evidence linking verified gradients to time to a validated target.

2 Related work

Wiesheu et al. (2024) combine spline-based rotor and stator shape optimization with engineering parameters for a nonlinear permanent-magnet synchronous motor. Their public implementation and saved outputs provide the reproduction target here. Their related study combines parameter and shape optimization with isogeometric analysis and compares against each alone (Wiesheu et al., 2025). We inherit the initial geometry, material table and excitation conventions; we replace the field discretization and compare two rotor–stator couplings. The resulting comparison is between numerical models, rather than a comparison with a measured physical motor.

Scotzniovsky et al. (2025) combine free-form deformation, differentiable mesh warping and electromagnetic analysis with adjoint-based derivatives. Merkel et al. (2021) use isogeometric shape derivatives and harmonic rotor–stator coupling to reduce electromotive-force harmonic distortion. Their objective is electromotive-force distortion, and their validation compares initial and optimized designs with a separate field solver. Here we examine torque standard deviation in a nonlinear reconstruction and test which interface model and excitation policy its gradients represent.

De Gersem and Weiland (2004) formulate harmonic weighting functions at the sliding interface of a finite-element machine model. Bontinck et al. (2018) use separate isogeometric subdomains and harmonic interface constraints; their exact circular geometry avoids polygonal air-gap error. Egger et al. (2022) derive torque from the energy of a harmonic mortar formulation and examine its numerical accuracy for ripple evaluation. Their linear-material cogging experiment varies the maximum transmitted order from 39 to 129 and compares against a refined order-200 reference. Thus torque sensitivity to interface order is already established; our study quantifies its effect on loaded nonlinear ripple, a saved optimized design, and design sensitivities. Bontinck et al. also analyze stability as the multiplier space grows relative to the field space, so adding orders requires compatible spatial resolution. We compare coupling torque with the gap-averaged stress expression of Arkkio (1987). These works motivate checking interface order, field resolution and torque evaluation together, rather than treating small mean-torque error as sufficient validation. The sensitivity of cogging-torque prediction to finite-element discretization and evaluation method was examined by Howe and Zhu (1992); Arkkio and Hannukainen (2012) study mesh selection for torque computation.

Ion et al. (2018) compare deterministic gradient-based optimization with stochastic search for electric devices, including a permanent-magnet motor, using geometry parametrizations that avoid remeshing. Section 7 extends that comparison with a coarse finite-element competitor and a common validation target. Its time-to-target measure follows data and performance profiles for benchmarking derivative-free solvers (Dolan and Moré, 2002; Moré and Wild, 2009), here with a fixed wall-time budget and censored targets.

Kemmetmüller et al. (2015) use nonlinear magnetic equivalent circuits for torque control, including saturation and nonsinusoidal behavior. Their control problem differs from shape optimization, but it demonstrates why saturation alone does not establish the need for a full-field model. Farshbaf Roomi et al. (2022) develop a waveform-targeting surrogate for electrical machine design, while Wen and Zahr (2023) adapt reduced models along a shape-optimization trajectory in fluid examples. These approaches motivate waveform-aware and adaptive baselines. Their published performance is not transferred to the motor studied here.

3 Motor model and reconstruction

3.1 Geometry and operating states

The full machine has six poles and 36 stator slots. We model a 60° sector, occupying $60° \le \theta \le 120°$ at zero rotation, with antiperiodic conditions on the radial boundaries. The rotor inner and outer radii are 16 and 44 mm; the stator inner and outer radii are 45 and 67.5 mm. Axial length is $L = 0.1$ m. The 1 mm air gap is split at the coupling radius $r_c = 44.5$ mm. Figure 1 shows the material-tagged reconstruction.

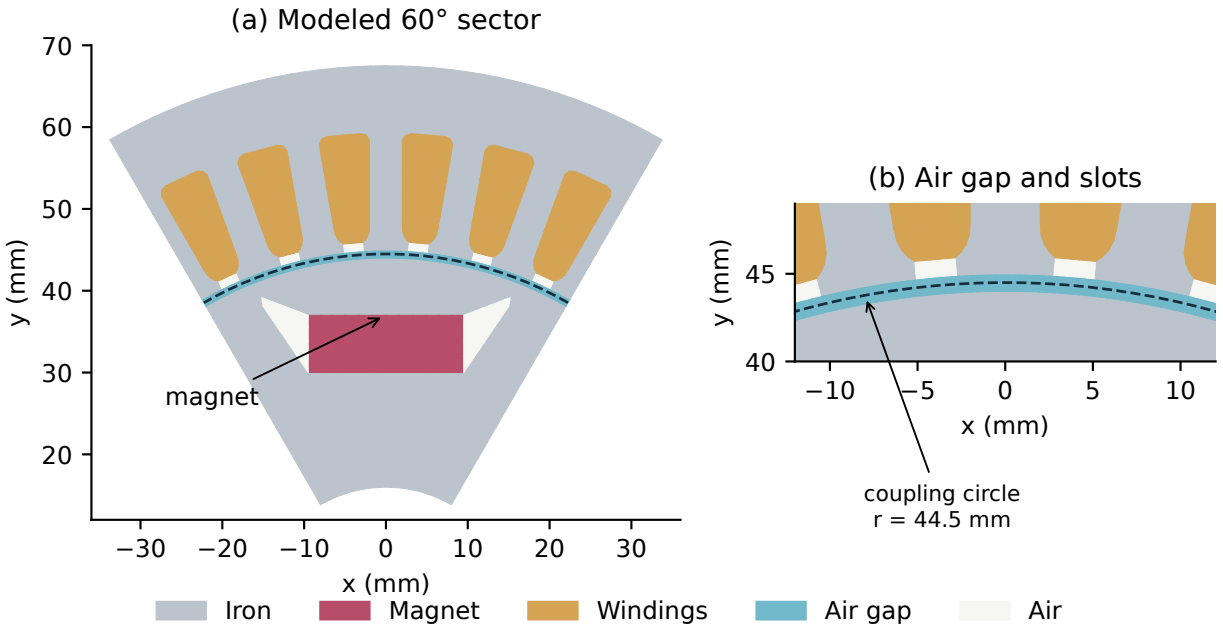


Figure 1: Initial-design geometry rendered directly from the committed coarsest mesh. The dashed line is the rotor–stator coupling circle. The right panel enlarges the gap and slot openings. Material boundaries are shown without the element edges; the geometry is not an optimized result.

The reconstruction uses the initial NURBS exports in the public snapshot. Their control points are stored with weights already multiplied in, and patch orientation must be accounted for when integrating areas. Two thin rotor patches described as air slits in source comments receive iron under the reference's material assignment; we follow that assignment. The mesh generator preserves the winding groups separately, since their two characteristic areas are approximately 36.29 and 31.22 mm^2.

At mechanical angle β, phase currents are

$$I_q(\beta) = aI_* \sin(\phi_0 + 3\beta + \delta_q), \qquad (\delta_U, \delta_V, \delta_W) = (0, -2\pi/3, 2\pi/3), \tag{1}$$

where $a \in \{0, 0.5, 1\}$, $I_* = 9.677657374\,\mathrm{A}$, and the electrical phase offset $\phi_0 = 0$ for the initial design. Angles in Eq. (1) are in radians. Each winding group carries current density $J_g = 12 s_g I_q / |\Omega_g|$, with its own area and prescribed sign $s_g \in \{-1, 1\}$. Using a common area would change the current density between groups by about 16%. We compare 20 distinct mechanical angles, $0°, 1°, \ldots, 19°$, excluding the duplicated endpoint in the source CSV. The reference's optimization grid contains the ten even angles; both grids are retained in the artifacts.

3.2 Nonlinear field equation

The two-dimensional magnetic vector potential $A = A_z$ determines the flux density $\mathbf{B} = (\partial_y A, -\partial_x A)$. Iron is described by an isotropic reluctivity $\nu(|\mathbf{B}|^2)$, where reluctivity is the inverse of permeability. For a test function v that satisfies the homogeneous constraints, the weak equation is

$$\int_\Omega \nu(|\nabla A|^2) \nabla A \cdot \nabla v \, \mathrm{d}\Omega = \int_\Omega J_z v \, \mathrm{d}\Omega + \int_{\Omega_m} \mathbf{M} \cdot (\partial_y v, -\partial_x v) \, \mathrm{d}\Omega. \tag{2}$$

Here $\mathbf{M}$ denotes the permanent-magnet source in $\mathbf{H} = \nu \mathbf{B} - \mathbf{M}$. In the magnet, $\mu_r = 1.05$ and $B_r = 1.4\,\mathrm{T}$, so its source magnitude is $H_c = B_r/(\mu_0 \mu_r)$, not B_r/μ_0. Its direction is $\pi/2$ in the rotor frame and rotates with the rotor. Air and windings have $\mu_r = 1$. The inner and outer circular boundaries have $A = 0$.

We use continuous piecewise-linear triangles. With nodal vector u, element gradients $g_i = \nabla \varphi_i$, $G = \nabla A$ and $s = G \cdot G$, the residual derivative on one element is

$$K^e_{ij} = |\Omega_e| \left[\nu(s) g_i \cdot g_j + 2\nu'(s)(g_i \cdot G)(g_j \cdot G) \right]. \tag{3}$$

The second term is required when the material changes with field strength. Newton steps use this consistent tangent, with damping and a stopping reference independent of the initial guess. Constant potential offsets, where allowed by the boundary conditions, do not change that reference. The solver reports whether it stopped at the requested tolerance or an estimated floating-point floor. All 500 baseline and sensitivity solves, and all 1,140 additional harmonic-coupling solves, stopped at the requested tolerance.

The iron law uses 47 tabulated M27 B–H points from 0 to 2.30 T. Monotonicity is imposed on $H(B)$, not on $\nu(B^2) = H(B)/B$. The tabulated H/B is non-monotone up to about 0.75 T. The two material tangent eigenvalues must satisfy

$$\nu > 0, \qquad \nu + 2s\nu' = \frac{\mathrm{d}H}{\mathrm{d}B} > 0. \tag{4}$$

The GEODE and reference interpolants agree to round-off between 0.05 and 2.25 T. GEODE uses an odd cubic near zero and a continuous slope into the vacuum-slope tail above 2.30 T. The reference uses a low-field clamp and a different last-knot slope. The maximum relative difference in reluctivity near zero is 6.66%; the last segment differs in H by at most 2.17×10^{-4} relatively. We test the torque effect of the low-field difference in Section 5.3.

3.3 Rotation and torque

At the modeled angles, the two circular traces have matching nodes after rotation. Our conforming interface identifies their potentials, including the antiperiodic sign on sector wrap-around. Writing the full nodal vector as $u = P\hat{u} + u_D$, the reduced residual and tangent are $\hat{R} = P^T R(P\hat{u} + u_D)$ and $\hat{K} = P^T K P$. The rotor geometry and magnet source rotate together; currents use the same mechanical angle in Eq. (1).

We average Maxwell stress over the air gap using the Arkkio expression (Arkkio, 1987)

$$T(\beta) = \frac{L}{\mu_0(r_2 - r_1)} \int_{\Omega_{\text{gap}}} r B_r B_\theta \, \mathrm{d}\Omega, \qquad (r_1, r_2) = (44, 45) \text{ mm}. \tag{5}$$

Positive torque acts counter-clockwise on the rotor. All torques in this paper include axial length and refer to the modeled sector. We apply no factor of six when comparing with the saved reference arrays, despite a per-length label appearing in the reference evaluation plots.

The alternative harmonic model imposes

$$\int_{\Gamma_c} (A_s - A_r) e_n \, \mathrm{d}\Gamma = 0, \qquad e_n \in \{\sin(n\theta), \cos(n\theta)\}, \quad n = 3, 9, \ldots, 45. \tag{6}$$

The 16 constraints match the reference's eight orders. Rotor and stator retain distinct trace unknowns. With constraint matrix C, Lagrange multipliers η and reduced potential $\hat{u}$, Newton's method uses the saddle tangent

$$\begin{bmatrix} K & C \\ C^T & 0 \end{bmatrix}. \tag{7}$$

The trace moments use the exact Fourier integrals of the piecewise-linear hats. Rotation enters their centers; the constrained energy derivative gives

$$T_c = -L\eta^T \frac{\partial (C^T \hat{u})}{\partial \beta}, \tag{8}$$

with the state held fixed in the partial derivative, consistent with Egger et al. (2022). Forward comparisons use Arkkio for both GEODE couplings to isolate the interface change; coupling torque is a separate check. The derivative study uses coupling torque for the harmonic model and Arkkio for the conforming model. The two objectives and their gradients are consequently comparisons of those specified model–estimator pairs.

The main harmonic runs retain antiperiodic trace-end pairing. An additional set frees those ends, matching the reference's treatment. Increasing the transmitted orders to every mode below the trace Nyquist limit reproduces the conforming Arkkio torques within 4.1×10^{-9} N m on L0 and 1.3×10^{-9} N m on L1. This checks the interface implementation without fitting any parameter to the saved torque.

4 Verification and experiment design

The nonlinear implementation is checked against linear assembly, residual finite differences, a manufactured solution and a saturated annulus with an independent radial field relation. Solver verification records are linked in the data-availability section; motor derivative checks appear in Section 6.

Sector tests compare the constrained solution with a full-machine solve, check the reduced tangent by finite differences, and exercise positive, negative and wrapped rotation steps. Clockwise and counter-clockwise triangle orderings give equivalent magnet sources and fields. The motor regression also detects changes in magnet strength, magnet direction, phase ordering and rotation direction.

The motor study uses four meshes, L0–L3, with 3,153, 9,937, 36,083 and 139,268 nodes. The 60° coupling arc has $120 \cdot 2^\ell$ segments at level ℓ, so integer-degree rotation is represented at every level. Each level contains 60 baseline solves: three current levels and 20 angles. A solution at one

angle initializes the next angle at the same current. We report the mean and population standard deviation,

$$\bar{T} = \frac{1}{N}\sum_{k=1}^{N} T_k, \qquad \sigma_T = \left[\frac{1}{N}\sum_{k=1}^{N}(T_k - \bar{T})^2\right]^{1/2}, \quad N = 20. \tag{9}$$

The standard deviation is the ripple metric used for the comparison. At zero current it measures cogging, the torque variation caused by the permanent magnets and slots without applied phase current.

For a scalar statistic q, the estimated L3 refinement interval is

$$q_3 \ \pm \ e_q, \qquad e_q = \max\{|q_2 - q_1|, |q_3 - q_2|\}. \tag{10}$$

We use the same construction pointwise for waveforms. This interval is a refinement diagnostic, not a proven error bound or a statistical confidence interval. It does not cover reference-model error or angular quadrature error. The loaded means change direction between the last two increments, so a last-increment estimate alone would be particularly weak.

5 Forward reproduction results

5.1 Coupling changes the ripple objective

Table 1 compares the same geometry, material and excitation with two interface models. The conforming L3 means are 0.15% and 0.16% low at half and full current. Harmonic coupling reduces the mean deviations to −0.002% and +0.051%, with both reference means inside the estimated refinement intervals. The loaded ripple deviations fall from +9.9% and +11.3% to +1.7% and +1.3%.

Table 1: L3 population standard deviation and mean torque, in N m per sector, on the 20-angle grid. The last column gives ripple deviation from the saved reference; n/a identifies the reference itself. The reference is subsequently refined in Section 5.4.

Current fraction	Model	Mean	Std	Std vs ref. (%)
0	Reference	0.00010	0.01071	n/a
0	Conforming	0.00001	0.00760	-29.1
0	Harmonic	0.00000	0.01209	+12.9
0.5	Reference	0.66208	0.03895	n/a
0.5	Conforming	0.66106	0.04279	+9.9
0.5	Harmonic	0.66206	0.03961	+1.7
1	Reference	1.44248	0.07935	n/a
1	Conforming	1.44023	0.08829	+11.3
1	Harmonic	1.44322	0.08036	+1.3

Figure 2 shows the improvement angle by angle. The largest full-current difference falls from 0.0272 to 0.0034 N m. The harmonic full-current ripple interval contains the saved reference; the saved half-current value lies approximately 5×10^{-5} N m below its interval. Mean agreement is therefore insufficient to characterize waveform accuracy.

The saved zero-current reference exposes a separate discretization effect. Harmonic coupling changes the cogging standard deviation from 29.1% below to 12.9% above the reference. Its waveform is close to a 1.13-fold scaling of the saved waveform, but the reference lies outside the estimated ripple interval. This motivates refining the reference itself before interpreting the discrepancy. The harmonic cogging standard deviations are 0.0101, 0.0119, 0.0121 and 0.0121 N m on L0–L3,

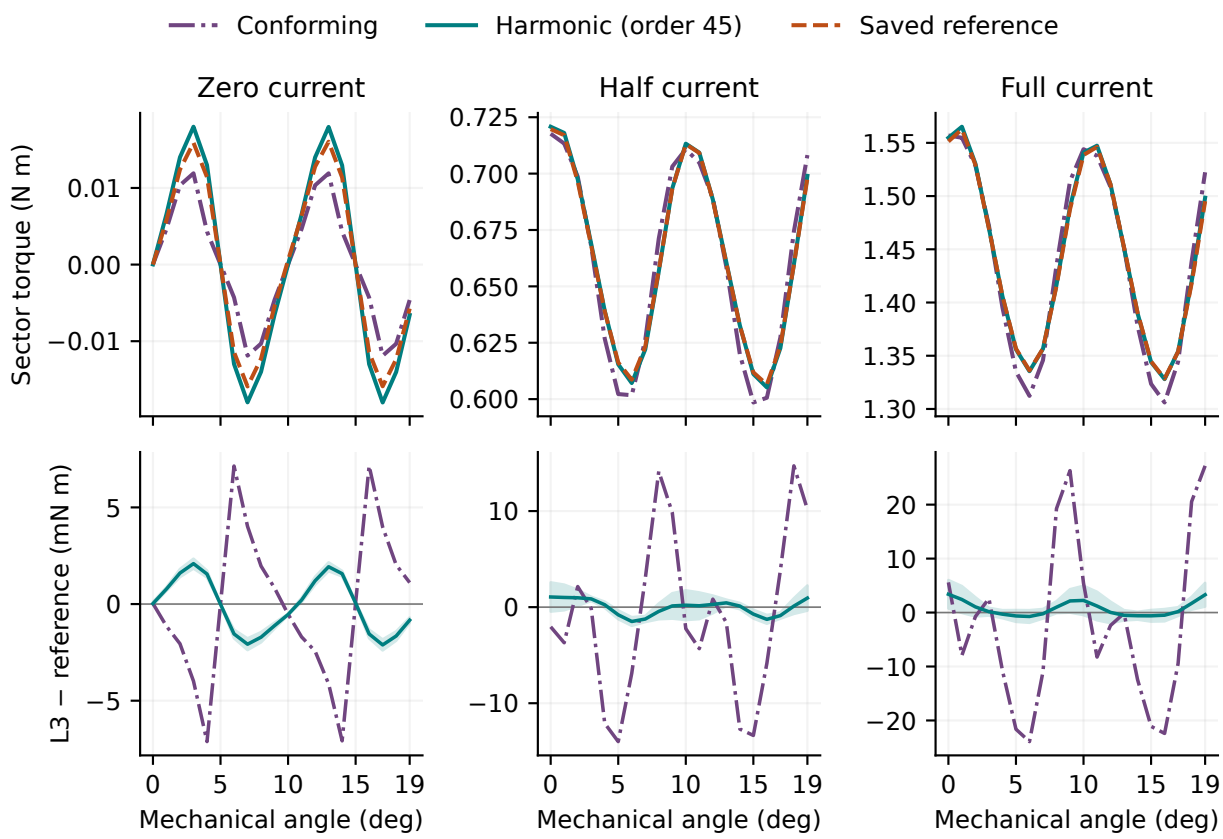


Figure 2: Top: L3 torque for conforming and order-45 harmonic coupling, compared with the saved reference. Bottom: each GEODE waveform minus the reference; shading is the harmonic model's pointwise estimated refinement interval from Eq. (10). Residual panels have independent scales. All GEODE curves use Arkkio torque, so the paired difference changes the coupling while holding the estimator fixed. The saved reference is coarse; Section 5.4 reports its refinement.

compared with 0.0107 N m in the reference. The independent reference refinement below tests this coarse-model explanation directly.

5.2 Spatial resolution and transmitted order

Spatial refinement at fixed order and order refinement on a fixed mesh answer different questions (Fig. 3). At L3, order 45 raises cogging standard deviation by about 59% and lowers loaded standard deviation by 7–9%, relative to the conforming values. At L1 and L2, order 189 approaches the conforming torque to within 1.1×10^{-4} N m. A spatially refined order-45 solution thus remains a solution of a truncated interface model. Matching that truncation improves reproduction of the published calculation; it does not establish which model better represents the physical motor.

Two additional controls narrow the explanation. At L3, coupling torque and the inner- and outer-half Arkkio estimates are each within 9×10^{-5} N m of full-band Arkkio in the order-45 solution. Freeing the coupling-arc ends changes torque by less than 5.4×10^{-5} N m. These effects are too small to explain the remaining cogging gap at the tested resolution.

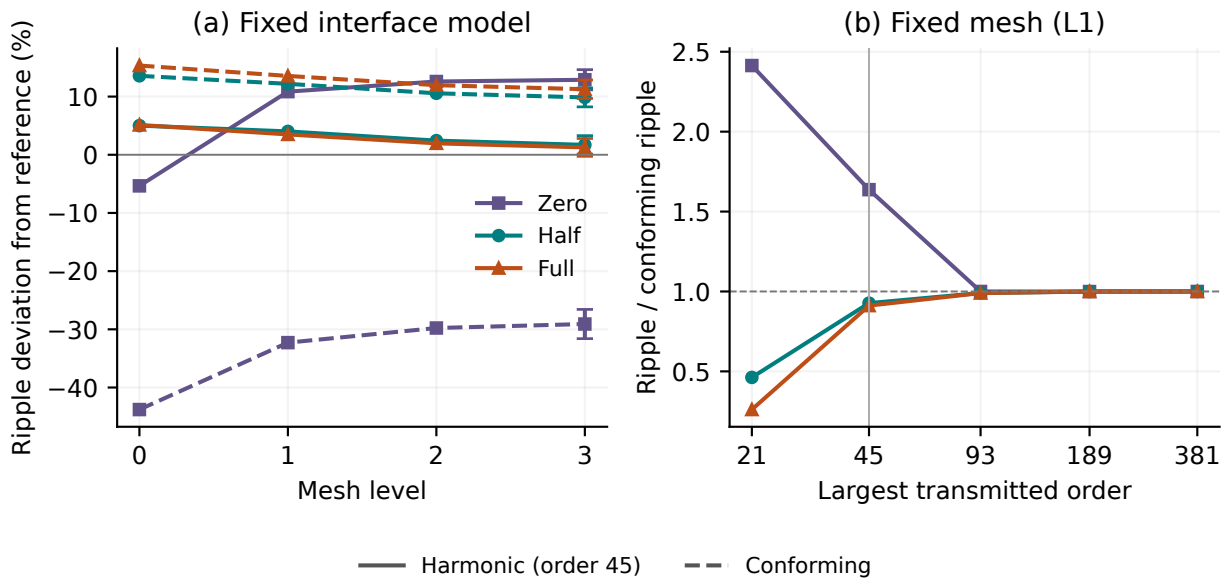


Figure 3: Left: ripple deviation from the reference under spatial refinement, with solid lines for harmonic order 45 and dashed lines for conforming coupling; L3 bars use Eq. (10). Right: order refinement on L1 with logarithmic order spacing, normalized by the conforming ripple at that same level. The saved reference uses order 45; see Section 5.4 for its refinement. Colors identify current, not coupling.

5.3 Sensitivity controls

The controlled changes in Table 2 are too small to explain the observed waveform gap at the tested settings. The near-zero material difference has particularly little effect on torque despite its 6.66% local reluctivity difference. This is a measured sensitivity of the initial design and does not establish insensitivity after optimization.

Table 2: Maximum absolute torque change relative to the baseline at the same mesh level, using conforming coupling and Arkkio torque. Entries summarize the recorded angle/current subsets.

Change	Levels	Max. $\lvert\Delta T\rvert$ (N m)
Reference low-field interpolation and clamp	L2, L3	2.1×10^{-6}
Relative Newton tolerance 10^{-12} vs 10^{-8}	L2; L3 full current	7.3×10^{-8}
Exact winding areas vs mesh areas	L2	3.9×10^{-4}
Inner or outer half of the Arkkio band	L3	7.3×10^{-5}

In the recorded estimator study, single-circle Maxwell-stress estimates vary more with the integration radius than the half-band Arkkio estimates. The paired coupling-torque and half-band checks above avoid selecting a favorable radius. A post-processing filter of the solved conforming field also gives non-monotone ripple changes across the tested cuts and currents; it does not replace a field solve with a truncated interface. The complete filter results are retained in the accompanying forward-study records.

5.4 Independent reference refinement

We rerun the public reference with its own degree-two isogeometric model, harmonic orders $3, 9, \ldots, 45$ and coupling torque, at 3, 6, 12 and 24 subdivisions in each patch direction. The 3×3 rerun first

reproduces all 60 saved torques with differences at most 9.8×10^{-13} N m. The run uses Octave 11.3.0, GeoPDEs revision `53ae5ca1` and nurbs 1.4.3; compatibility edits, source hashes and raw records are retained. The corresponding field degrees of freedom are 4,079, 12,518, 43,220 and 159,920.

The reference cogging standard deviation changes from 0.01071 to 0.01220, 0.01212 and 0.01208 N m (Fig. 4a). GEODE L3 at order 45 gives 0.01209 N m, a 0.10% difference from the finest reference. The maximum pointwise discrepancy drops from 2.1×10^{-3} to 3.4×10^{-5} N m. At half and full current, the finest reference standard deviations are 0.03947 and 0.08014 N m; GEODE is 0.35% and 0.27% higher (panel b). All five finest-reference statistics, the three standard deviations and two loaded means, lie within GEODE's L3 refinement intervals. The coarse saved reference therefore accounts for the cogging mismatch in Section 5.1 at the matched interface order.

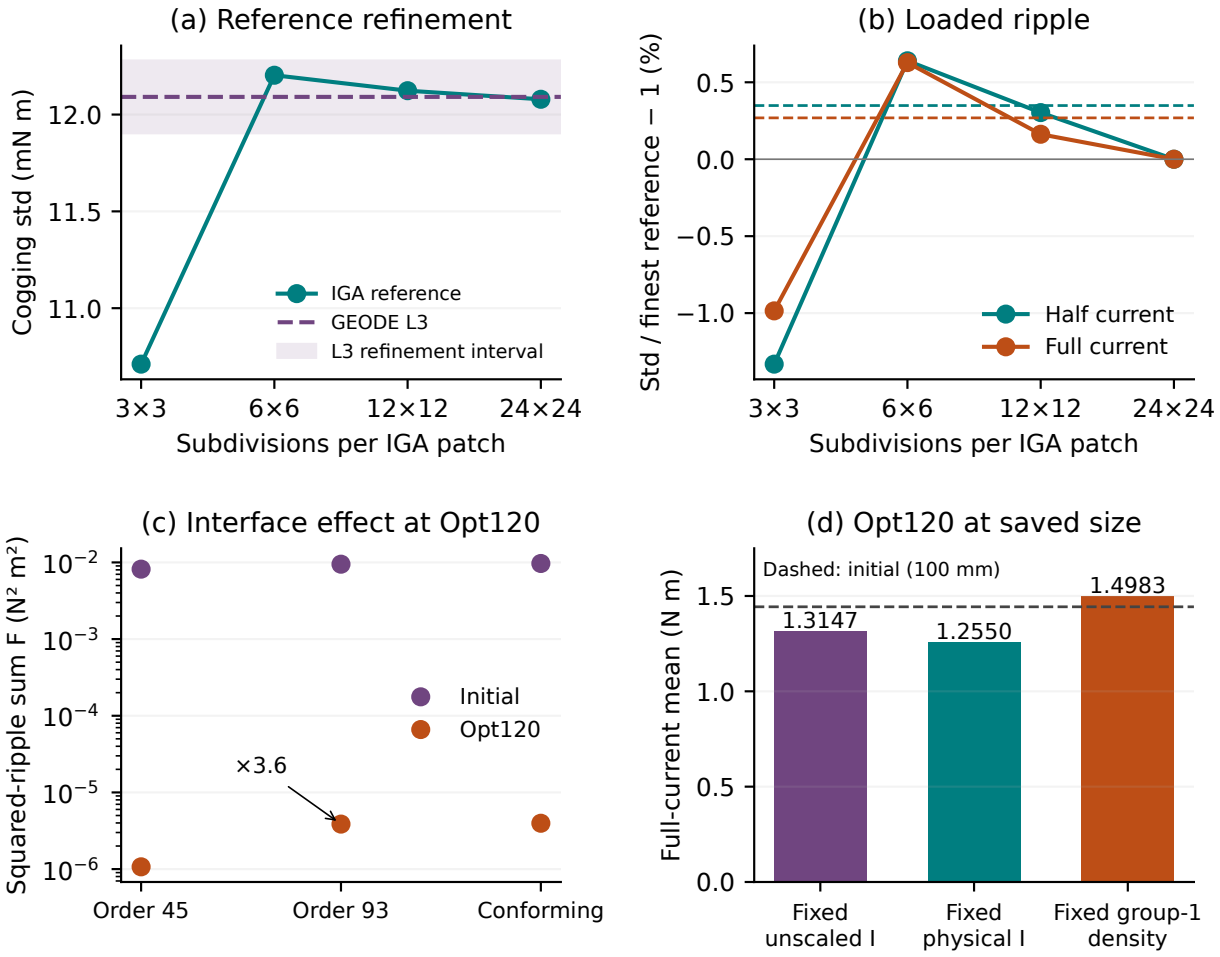


Figure 4: Reference and candidate validation from raw torque records. (a) Independent IGA reference cogging refinement at order 45; the dashed line is GEODE L3 and its shaded band uses Eq. (10). (b) Loaded standard deviation relative to the finest reference at each current; circles are IGA, horizontal lines are GEODE L3. The finest IGA point is zero by construction. Panels (a,b) use the initial geometry and 20 angles; GEODE uses Arkkio, with coupling torque checked separately. (c) Initial and Opt120 squared ripple sums at L3 on fresh meshes and 20 angles, under fixed unscaled current. Harmonic values use coupling torque; conforming values use outer-band Arkkio. (d) Full-current mean at order 45 for Opt120 under three current rules at its saved size; the dashed line is the initial mean at the initial size. All candidate values use coupling torque.

This agreement still concerns finite discretizations. The last cogging increments approximately

halve; extending that trend would put a limit near 0.0120 N m, an estimate without a bound. The last two half-current standard-deviation increments are both about −0.3% and do not shrink; we therefore leave its remaining error unestimated. The 0.10% cogging agreement is a comparison of finite-resolution values, not an accuracy bound; neither refinement sequence is fully converged. Tightening the reference Newton stopping threshold from 10^{-6} to 10^{-14} changes the checked cogging torques by at most 4.4×10^{-8} N m. GEODE retains straight-sided geometry, and the high-field extrapolation of the iron law has not been varied in a paired torque experiment. Neither cross-code agreement nor refinement proves physical accuracy for the machine.

5.5 Validation of the saved optimized design

The saved reference design at iteration 120, denoted Opt120, includes 12 scalar parameters and ten grouped surface-control offsets. We evaluate its full exported geometry on freshly generated meshes; simply moving the finest initial mesh produces four inverted triangles. Its inner air-gap band no longer has the annular area assumed by the full-band Arkkio functional, so harmonic results here use coupling torque. Conforming results use only outer-band Arkkio on the fresh exported mesh. On the same L3, order-45 Opt120 field at full current, the outer-band and coupling estimates differ by about 1.3×10^{-4} N m in mean and 1.1×10^{-5} N m in standard deviation. This paired check supports the estimator substitution at this resolution. This estimator restriction is specific to these geometries; it does not validate outer-band Arkkio on arbitrary moved meshes or stator offsets.

Define $F = \sum_{a \in \{0,0.5,1\}} \sigma_T(a)^2$, in $\mathrm{N^2\,m^2}$, on the 20-angle grid. With the evaluation current held fixed in the unscaled geometry, raising the transmitted order from 45 to 93 multiplies Opt120's F by 3.6, from 1.07×10^{-6} to 3.86×10^{-6} (Fig. 4c). The initial design's F rises by 16%. Its ratio to Opt120 is consequently 7,631 at order 45, 2,454 at order 93 and 2,444 for conforming coupling. The full-current Opt120 standard deviation rises from 0.00066 to 0.00164 N m. The large ripple reduction survives these checks, but its magnitude depends strongly on the interface model. Figure 5 shows the full and mean-subtracted waveforms.

Between L2 and L3, changes in F, normalized by L3, reach 1.36% for the initial design and 2.71% for Opt120 across the three models on the exported mesh family; these increments are not error bounds. For Opt120 at order 45, the successive L1–L2 and L2–L3 increases in F are 3.05×10^{-8} and 2.90×10^{-8} $\mathrm{N^2\,m^2}$, so they barely shrink and do not establish an asymptotic rate. An alternate freshly constructed mesh family changes by 3.22% for Opt120 from L2 to L3. The 2.79×10^{-6} $\mathrm{N^2\,m^2}$ interface effect is about 96 times the last export-mesh increment, despite this unresolved remainder.

A possible explanation is that optimization against order 45 favors cancellation within that truncated model. This is a conjecture, not an ablation result: increasing the order raises full-current ripple but lowers Opt120 cogging standard deviation from 0.00061 to 0.00035 N m.

Current policy also changes feasibility (panel d). At L3 and order 45, the initial full-current mean is 1.4433 N m. Opt120 gives 1.3147 N m under fixed *unscaled* current, 1.2550 N m under fixed physical current, and 1.4983 N m under the reference's fixed group-1 density rule. The first two are 8.9% and 13.0% below the initial mean; the density-rule value is 3.8% above it. These policy comparisons use the same Opt120 geometry, but comparisons with the initial motor also change size: Opt120 shortens the stack from 100 to 80 mm and has radial scale $s = 1.0477$, unscaled outer diameter 130.293 mm and operating angle −11.646° (Table 8). Every Opt120 torque carries the explicit size factor $s^2 L/(0.1\,\mathrm{m}) = 0.878175$. Removing both size factors from the fixed-unscaled-current mean gives 1.4971 N m, 3.7% above the initial mean. Applying the same normalization to fixed physical current gives 1.4291 N m, 1.0% below the initial mean; the density-rule value becomes 1.7061 N m, 18.2% above it. Normalizing only length to 100 mm under fixed unscaled current gives 1.6434 N m (+13.9%), retaining the radial factor $s^2 = 1.098$. These algebraic normalizations separate the size

factors from the current rule; they are not reoptimized designs. Opt120 is outside the prospective fixed-size, fixed-physical-current problem.

A separate reproduction comparison uses the saved reference current at each design, rather than fixed unscaled current. At Opt120, GEODE's order-45 full-current mean is 1.4986 N m versus 1.5000 N m in the saved arrays, but its standard deviation is 3.53×10^{-4} versus 1.76×10^{-4} N m, a factor of 2.0. The half-current values are 5.53×10^{-4} and 4.37×10^{-4} N m. The largest pointwise difference across the two loaded currents is about 2.1×10^{-3} N m. With this matched current policy and zero-current values included (zero current is rule-independent and taken from the fixed-current record), the initial-to-Opt120 objective ratios are 10,170 for GEODE and 11,982 for the saved reference. The 7,631 ratio above instead holds unscaled current fixed. Reference refinement was run only at the initial design, so the cause of the Opt120 discrepancy remains unresolved.

Angular sampling becomes more consequential at this low ripple. Relative to the 40-angle grid, the ten-angle Opt120 objective reads about 1.05% high at order 45 and 5.90% low at order 93 on the fresh L3 mesh; the conforming value reads 17.1% low. The accompanying records retain shifted 20-angle grids and withheld currents 0.25 and 0.75. Candidate validation must check these effects at the candidate, since initial-design agreement alone would miss them. All 3,240 solve rows in the 18 successful records used for these candidate comparisons stopped at tolerance.

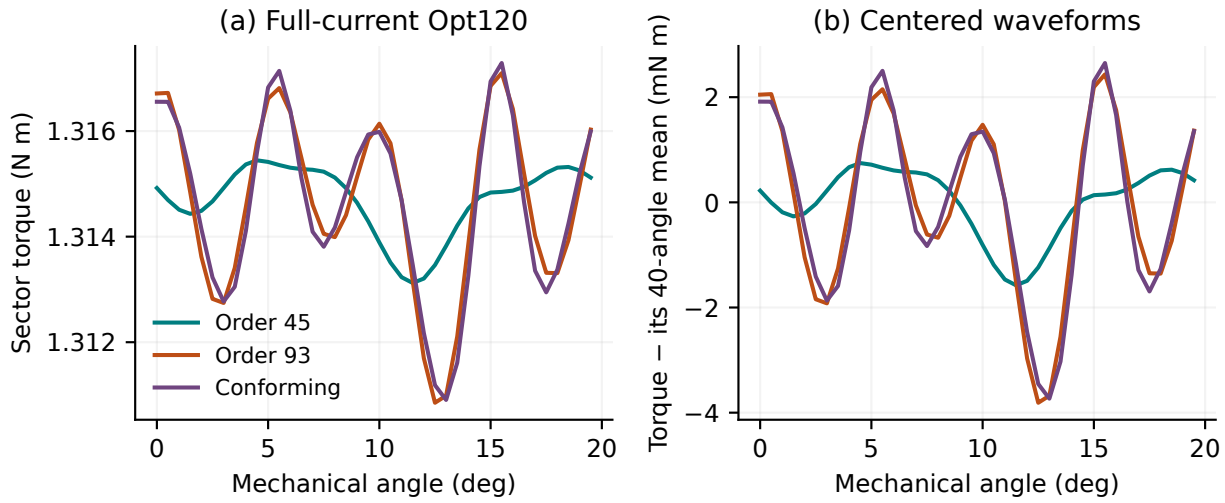


Figure 5: Opt120 full-current torque on fresh L3 meshes at 40 angles, with fixed unscaled current. Harmonic orders 45 and 93 use coupling torque; conforming coupling uses outer-band Arkkio. Left: full torque, retaining each model's mean. Right: each curve minus its own 40-angle mean. The interface change mainly alters the resolved ripple.

6 Design-gradient verification

6.1 Differentiated model and excitation policy

For a design vector p, the converged state z contains the reduced potential and, for harmonic coupling, its multipliers. Writing the full constrained residual as $R(z, p) = 0$, the discrete adjoint is

$$R_z^T \psi = F_z^T, \qquad \frac{\mathrm{d}F}{\mathrm{d}p} = F_p - \psi^T R_p. \tag{11}$$

We differentiate mesh coordinates, element terms, magnet sources, current loads, harmonic constraints and the torque functional. The explicit coordinate derivative of Arkkio is included

for conforming coupling. Fixed-topology mesh motion $X(p)$ supplies the chain rule $\mathrm{d}F/\,\mathrm{d}p_k = \langle \mathrm{d}F/\,\mathrm{d}X, \partial X/\partial p_k \rangle$ for geometric parameters, with direct terms added where needed.

The 12 parameters comprise nine geometric lengths (MAG, MH, MW, SW1–SW4, SD1 and Sr1), axial length, operating angle and radial scale. The two designs are the initial parameters and an interior point halfway toward the saved optimized scalar parameters (Appendix B). This verifies a local scalar parametrization. The reference's optimized geometry also uses grouped surface-control displacements; reproducing its 12 scalar values alone does not reproduce that geometry. The present gradient comparison does not include derivatives of those additional groups or a new optimized motor.

In this scalar derivative study, fixed current means fixed peak current in the unscaled geometry, so $J_g = 12 s_g I_q / |\Omega_g(X)|$ changes when a slot changes area. For comparison we implement the reference optimizer's rule, denoted *fixed group-1 density*: $I_* \propto |\Omega_1(X)| s$, where s is radial scale, with the proportionality constant fixed at the initial design. The unscaled geometry is evaluated and torque is multiplied by s^2, following the reference. Both rules agree at the initial geometry but have different derivatives. Group-2 density is not held fixed because the ratio of the two winding areas can change. At L1 (harmonic coupling, coupling torque, full current), the mean-torque derivative with respect to SW1 is $+1.68\,\mathrm{N\,m/m}$ under fixed current and $-220.85\,\mathrm{N\,m/m}$ under fixed group-1 density (Fig. 6). SW1 is the stator-tooth-width parameter. The sign change comes from the additional winding-area/current term: a wider tooth narrows the slot and lowers current under fixed group-1 density. At this state the policy changes seven of the ten geometric/scale derivatives and reverses the mean-torque signs for SW1, SW4 and SD1. Rotor parameters MAG, MH and MW leave winding area unchanged and therefore give identical derivatives under the two rules. The prospective fixed-size problem sets radial scale to one, making fixed physical and unscaled current coincide. A future optimization comparison must hold this policy fixed across methods.

6.2 Finite differences and Taylor remainders

The derivative objectives are mean torque and population standard deviation on the ten even mechanical angles, matching the reference's optimization grid. Central differences move the same mesh at $p \pm h e_k$; step sweeps separate truncation from floating-point and nonlinear-solve error. We also check the first-order remainder $|F(p + h e_k) - F(p) - h\,\mathrm{d}F/\,\mathrm{d}p_k|$ for quadratic decay.

Across initial and perturbed designs, L0–L2 and the recorded loaded current/rule subsets, there are 800 objective–parameter records. They pool harmonic/coupling-torque and conforming/Arkkio evaluations. The fixed-density MAG, MH and MW records repeat 96 fixed-current comparisons byte-for-byte, leaving 704 distinct comparisons. Of these, 688 have a best-step relative error at most 2.8×10^{-7}; the maximum is 1.07×10^{-5} for a small Sr1 ripple derivative. This is a best-step verification statistic, not the accuracy of a finite-difference step chosen in advance. Figure 6 includes the 704 distinct cases and an explicit Taylor sweep. The artifact retains every step, rejected displacement and relative error. The recorded subsets do not form a full factorial study.

After removing duplicates and the analytic axial-length and radial-scale cases, 530 of 592 loaded comparisons have an anchored quadratic-remainder window spanning at least four decades in step size; 388 span at least five. The window is the contiguous run where remainder$/h^2$ stays within a factor of two of its value at $h = 10^{-5}\,\mathrm{m}$ (or 10^{-2} degrees for operating angle). Smaller gradients reach round-off sooner. An anchored-window diagnostic fails for the initial-design, full-current conforming/fixed-density SW1 ripple cases at L0 and L1: at L0 the remainder changes sign near the anchor, whereas at L1 remainder$/h^2$ is still drifting there. Their widest quadratic windows span two decades. The nearly cancelling zero-current mean is a weaker check: its best relative errors reach 7.0×10^{-4}. Cogging standard-deviation errors are at most 8.9×10^{-7}. Thus the loaded agreement does not describe every zero-current derivative.

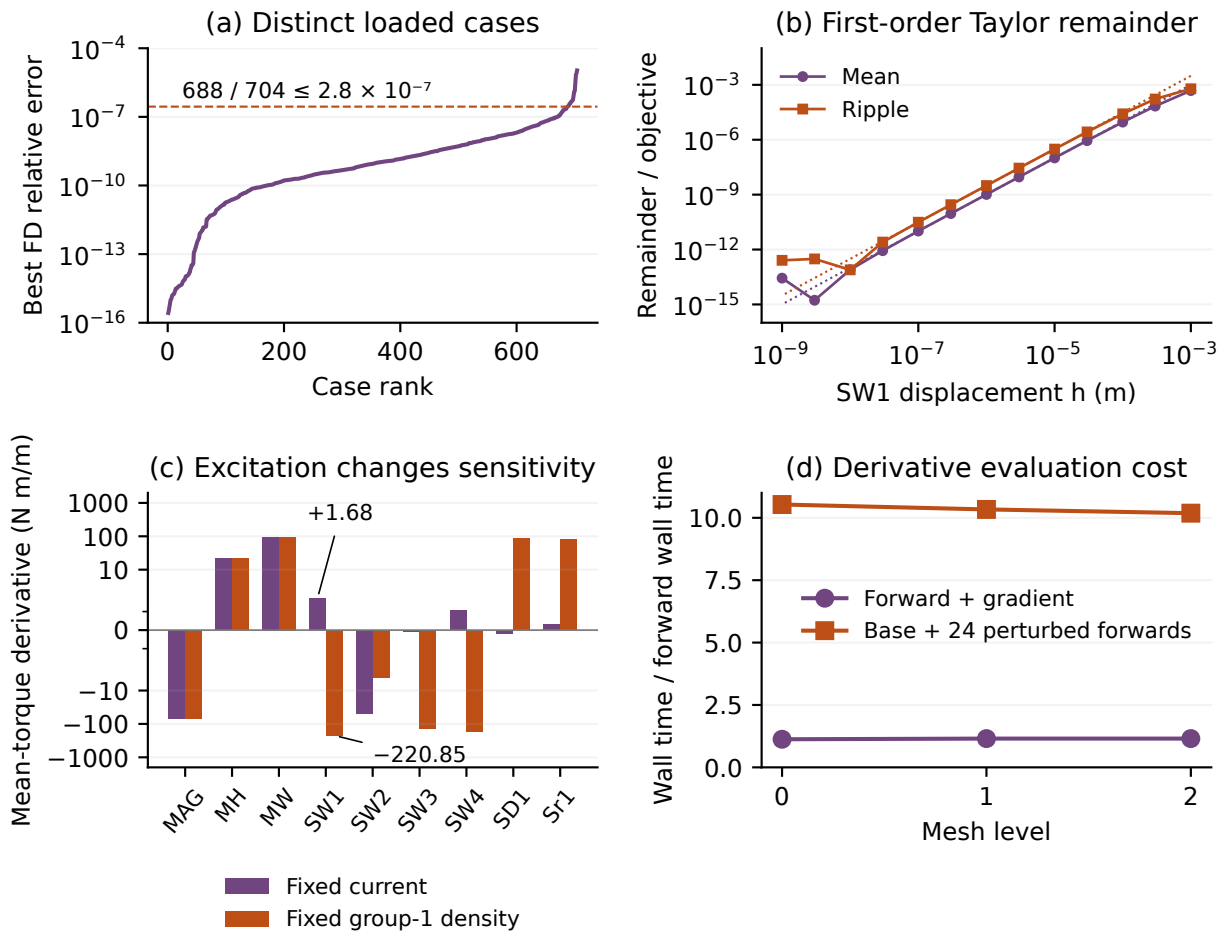


Figure 6: Discrete gradient evidence. (a) The 704 distinct loaded best-step finite-difference errors, sorted, pooling harmonic/coupling-torque and conforming/Arkkio objective–parameter cases. (b) SW1 Taylor remainders at L1, initial design, full current, harmonic coupling and coupling torque, fixed current; the plotted remainders are divided by $|F(p)|$, and dotted h^2 guides are anchored at $h = 10^{-5}$ m. (c) Mean-torque sensitivities for the nine geometric parameters under two current rules at that same state; the vertical axis is symmetric-log, linear within ± 2 N m/m. (d) Recorded cost ratios for harmonic coupling at tolerance 10^{-14}; gradients include the forward evaluation, and FD includes its shared base forward. L0/L1 use medians of three runs, L2 a single run. These timing data were collected under uncontrolled host load.

Discrete derivative accuracy also does not imply mesh convergence. In the harmonic model, SW3 mean-torque sensitivities change sign between coarse levels and their increments shrink with refinement; ripple gradients converge more slowly than most mean gradients. The conforming/Arkkio SW2 mean sensitivity is non-monotone across L0–L3 (-46.5, -41.3, -44.4, -47.2 N m/m), whereas harmonic/coupling torque approaches -47.1 from -49.8 N m/m. Fresh-remesh checks of MW, SW1 and SW2 are limited at small steps by changes in connectivity. The L2 MW mean-derivative fit has relative gap 2.6×10^{-6}, but individual remeshed steps at $h \geq 10^{-4}$ m scatter by about 6×10^{-4}. The largest L2 full-current remeshed-versus-moved fitted discrepancy is 3.8% for the SW1 ripple derivative. These checks support the discrete chain rule while leaving refinement of candidate designs necessary.

6.3 Model dependence of verified sensitivities

The two model–estimator pairs give different ripple sensitivities even at the same design (Fig. 7). At L2 and full fixed current, the MAG ripple derivative is 1.28 N m/m for harmonic/coupling torque versus 1.94 for conforming/Arkkio at the initial design. At the perturbed design the values are 0.75 and 1.87 N m/m. These differences persist under mesh refinement. At L3 the MAG pairs are 1.22 versus 1.87 at the initial design and 0.57 versus 1.67 N m/m at the perturbed design. In particular, the latter harmonic component changes appreciably even though the box-scaled vector changes by only 1.19% of its L2 norm; the norm itself decreases by 0.19%. Mean-torque sensitivities agree more closely in the box-scaled comparison: the Euclidean norm of the vector difference, divided by the conforming vector norm, is 0.58% and 0.66% for the mean at the initial and perturbed designs, versus 6.7% and 10.2% for ripple. The norm weights the largest components most heavily: the initial SW2 and SW3 mean components differ by 6.5% and 16.5% relative to conforming values. At L3, the ripple gaps remain 6.58% and 10.25%, while each model changes by 0.46–1.19% from L2 to L3. The mean gaps fall to 0.16% and 0.26%; the conforming model itself changes by 0.52–0.54%. The L2 mean gap is therefore comparable to spatial refinement, whereas the ripple gap persists beyond the measured refinement increments. Because the estimator changes with the model, this experiment measures their combined effect; it does not isolate an interface-only derivative error or establish which gradient is closer to the physical machine.

6.4 Recorded derivative cost

The recorded timings use an Apple M3 Ultra with 28 cores, default solver threading, and concurrent host load averaging 14–24. The effective thread count was not pinned. A forward evaluation covers ten angles, starts from zero, and warm-starts successive angles. The full gradient includes that forward, ten torque adjoints, mesh derivatives and contractions for all 12 parameters and both aggregate objectives. Both aggregates reuse the per-angle torque derivatives. Central differences use 24 forward sweeps, each warm-started from the base states, at step $h = 10^{-6}$ in each parameter's stored unit (m, degrees or dimensionless scale). We add the shared base forward to the recorded 24-sweep FD time for a common starting point. These derivative timings are a separate workload from the 1,640 forward validation solves.

Across L0–L2, both couplings and tolerances 10^{-8} and 10^{-14}, the full gradient costs 1.13–1.20 forwards, whereas central differences cost 8.2–10.5 including the shared base solve. Table 3 gives the harmonic tight-tolerance subset. Shape-map setup is recorded separately and must be included in a complete optimization budget. The cost ratio concerns derivative evaluation on one loaded host; Section 7 measures the charged cost of whole optimization runs and their time to a validated target.

Table 3: Recorded harmonic-model cost at relative tolerance 10^{-14}, initial design and full current. Times are seconds for ten-angle sweeps; FD includes the shared base sweep, and setup is a one-time shape-map cost.

Mesh	Forward	Forward + gradient	Base + FD	Setup
L0	0.664	0.748	6.997	0.011
L1	3.174	3.658	32.811	0.041
L2	12.702	14.644	129.390	0.222

7 Matched optimization study

The preceding sections establish that the gradients are accurate for the model they differentiate. This section tests whether they shorten the time needed to reach a common validated ripple target. The protocol and all of its amendments were fixed in the preregistration before any main-run

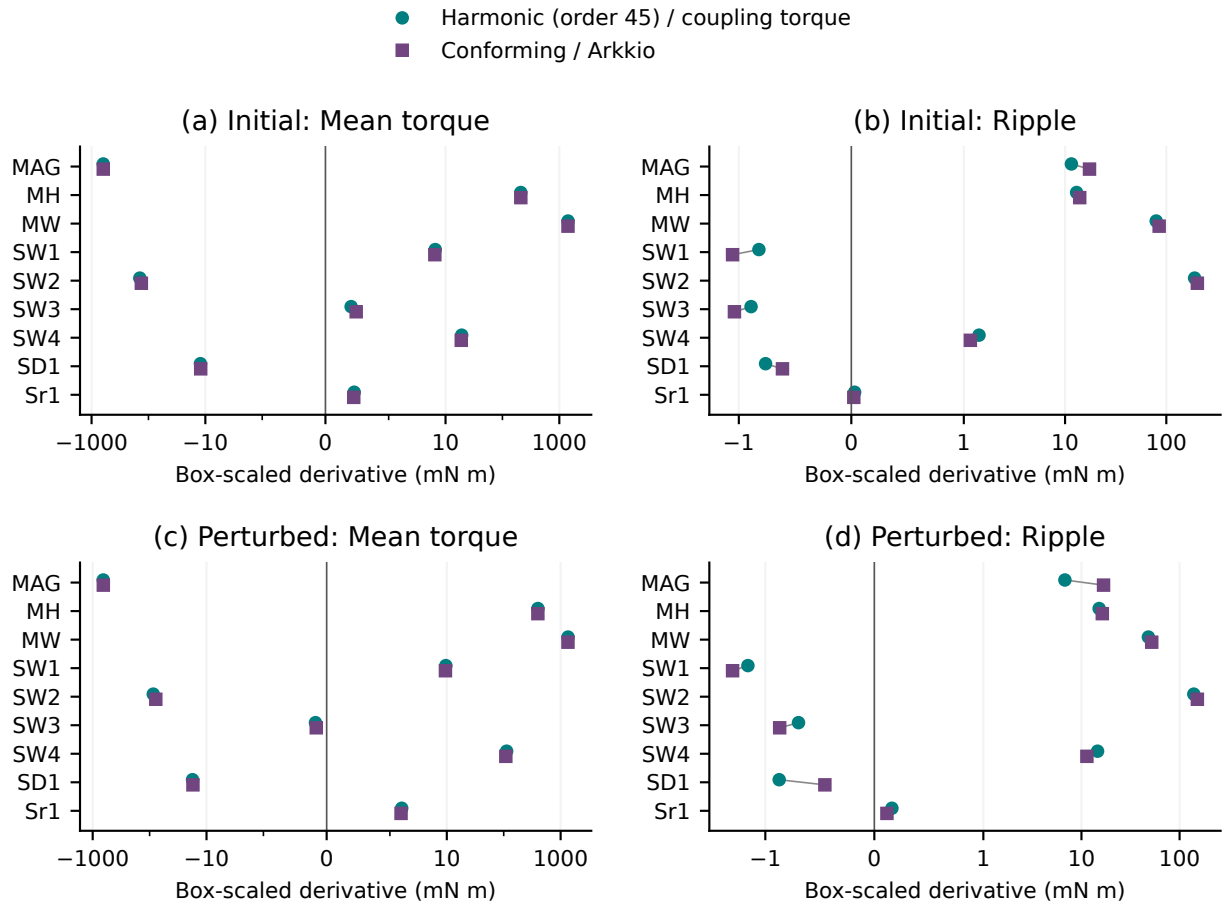


Figure 7: Full fixed-current sensitivities at L2, scaled by each geometric parameter's reference box width w_k. Top: initial design; bottom: perturbed design. L3 MAG values and refinement changes are given in the text. Left: mean torque; right: standard deviation. The plotted quantity $w_k \partial q / \partial p_k$ is a local sensitivity, not a finite change across the box. Coupling and torque estimator both differ between the two series. Gray segments join the two estimates for each parameter. Horizontal axes are symmetric-log, linear within ± 1 mN m. All nine widths, in mm, are 9, 10, 15, 4, 3, 1, 18, 70 and 1.5 in the displayed order.

outcome was observed. One later deviation, in how the final refinement was run, was adopted after the main runs had finished and is described with the refinement in Section 7.2; it changes no selection rule, check or target.

7.1 Protocol

Methods. Five methods form an optimizer-by-resolution comparison plus one surrogate (Table 4). A and E are safeguarded SLSQP (Kraft, 1988) using the field adjoints of Section 6; B and D are COBYQA (Ragonneau, 2022) with $2n + 1$ interpolation points and unit-box trust radii 0.1 to 10^{-6}, where n is the number of free parameters. Both run through SciPy 1.18.1 (Virtanen et al., 2020). C is an empirical cubic radial-basis trust-region surrogate fitted to L1 evaluations. A/B and E/D vary the optimizer at fixed training resolution, and A/E and B/D vary resolution at fixed optimizer. The optimizer contrast includes differences in search strategy, so it does not isolate gradient access as the only causal difference. Training uses order-45 coupling torque on 20 angles at current fractions 0, $\frac{1}{2}$ and 1 of the fixed peak current 9.6777 A. L0 and L1 differ only in spatial resolution, and neither

is fitted or corrected against validation. Time to a fixed target, with unattained targets kept as censored, follows the data- and performance-profile practice for benchmarking optimization solvers (Dolan and Moré, 2002; Moré and Wild, 2009). The preregistration also named a fixed group-1 current-density rule as a labelled control. It was not run in this study, which uses the fixed physical current throughout; Section 6 reports that rule's effect on sensitivities.

Table 4: Methods. D and E solve the coarse nonlinear field equations; they are neither fitted magnetic circuits nor reduced-basis models.

Method	Optimizer or search model	Training mesh
A	Safeguarded SLSQP, field adjoints	L1
B	COBYQA, forward evaluations	L1
C	Empirical cubic-RBF trust region	L1
D	COBYQA, forward evaluations	L0
E	Safeguarded SLSQP, field adjoints	L0

Design spaces and constraints. Space A has eight free geometric lengths: MAG, MH, MW, SW1–SW4 and Sr1 (Appendix B), that is, the rotor iron depth, magnet height and width, four stator tooth and slot dimensions, and the slot-bottom fillet radius. Space B adds ten grouped offsets acting on 55 surface control points. Stack length, radial scale, stator outer diameter and operating angle are fixed, so the saved Opt120 design lies outside both spaces. The two spaces also use different discrete mesh-motion maps, so their contrast does not isolate the ten added offsets. Each training model $r \in \{\mathrm{L0}, \mathrm{L1}\}$ minimizes $F_r = \sum_c \sigma_{r,c}^2$ over the three training currents, subject to $\mu_{r,1} \geq 1.0025\,\mu_{r,1}(\theta_0)$ for its own full-current mean and to normalized construction margins $c_i - 0.05 \geq 0$. Unbuildable or mesh-infeasible candidates are refused: SLSQP restores toward the last admissible point and restarts, COBYQA receives a NaN barrier, and C skips the point. A moved mesh with an inverted triangle or a minimum angle below 5° is repaired by up to four charged remeshing attempts. Because refinement cannot remove a limiting geometric corner, the 5° policy also acts as an implicit geometric constraint. Space B's slot-opening channel wall is not guarded and can become very thin at extreme offsets; that risk is handled by repair and refusal, not by a wall-thickness constraint.

Common validation and credit. Every method is judged on the same order-93, L2, 20-angle validation model. Its initial-design anchor is $F_0 = 9.6034 \times 10^{-3}\,(\mathrm{N\,m})^2$ with full-current mean $\mu_0 = 1.44084\,\mathrm{N\,m}$. The targets are $T_{50} = 0.5F_0$, $T_{90} = 0.1F_0$ and $T_{99} = 0.01F_0$. A validated design attains a target only if

$$F_{\mathrm{val}} + \delta F \leq T, \qquad \delta F = \max\{0.10F_{\mathrm{val}},\, 2.5 \times 10^{-7}\,(\mathrm{N\,m})^2\}, \tag{12}$$

its validated full-current mean is at least $1.0025\,\mu_0$, its construction margins and mesh checks pass, and its validation completes within budget. The allowance δF is an empirical margin, not a bound on continuum error. Optimizer success flags confer no credit. Changed incumbents are validated as they appear, after the initial interpolation set for B, C and D.

Budget and timing. Cost is wall time, in units of the median cold validation forward measured on the same host at the same concurrency. Each run receives 120 units, with a 60-unit checkpoint that counts only validations completed by then. The charge covers evaluator start-up, geometry checks, mesh motion and repair, training solves, adjoints, optimizer overhead, surrogate fitting and validation; nothing is subtracted as overhead. L0 receives no discount.

Starts and execution. Each space uses θ_0 and four Latin-hypercube starts (seeds 1–4), with the scalar coordinates matched across spaces. A start was eligible only if it was buildable and admissible, and its moved reference mesh met the mesh policy without repair, in both spaces at L0 and L1. The starts therefore favor moderate deformations and are not uniform samples of the design box. All starts were frozen before any main outcome. Every method runs from every start twice, giving 100 runs. Within a start, repeat 1 rotates the method order and repeat 2 reverses it.

The session ran on one AWS c7a.4xlarge host (x86-64; NumPy 2.5.4 and SciPy 1.18.1 with OpenBLAS) with 15 concurrent single-threaded processes. Before calibration, the host recomputed the initial-design validation waveforms in both spaces. F_0 agreed with the canonical anchor to 2.4×10^{-13} relative, within the preregistered 10^{-9} tolerance, and the two spaces agreed bit for bit. The calibrated unit was 70.94 s, the median of 45 concurrent cold validations. All 100 runs completed in one session with no recovery, deferral or contamination: the one-minute load average peaked at 21.1, below the contamination threshold of $1.5N = 22.5$. The in-run cost totals 109 CPU-hours. Of the 100 runs, 38 stopped on reaching T_{99}, 46 on optimizer convergence and 13 at the budget. The other three are SLSQP runs (A) in space A whose solver stopped without converging: both repeats from LHS 2, after about 43 units, and repeat 1 from LHS 3, after 119 units.

7.2 Results

Repeats. The two repeats of each run follow the same optimization path. In all 50 pairs, the sequences of validated objectives agree exactly up to the shorter run's length, and they differ in length only when the budget or a solver stop cuts one repeat earlier. The repeats therefore measure timing noise, not independent outcomes, and the comparison is a paired case study over five starts. Over the 111 target crossings attained in both repeats, the time differs by a median of 2.1% and at most 12.4% relative to repeat 1. Repeat-1 times are reported below.

Attained quality. Figure 8 shows the best credited value of $(F + \delta F)/F_0$ for every start, method and space (repeat 1). In space A, no run goes below $F/F_0 = 0.043$ and none reaches T_{99}. The best designs of A, B, D and E all reach $0.043F_0$, and C's reaches $0.049F_0$; in Fig. 8, which includes the allowance, the level appears near 0.048. All five lie within 0.17% above the full-current mean floor, so the mean constraint is active there: the level shared by A, B, D and E is the constrained optimum they found in the eight-parameter space, not a property of any one method. In space A the start mattered more than the method. From LHS 4, A and B stop near $0.09F_0$, D reaches $0.044F_0$, and C and E never produce a credited validation, in either repeat. In space B, each method's best design reaches 0.004–$0.009F_0$, and every method except C reaches T_{99} from at least four of five starts.

Time to target. Table 5 and Figure 9 give the charged time to each target, per start. In repeat 1, only three crossings occur after the 60-unit checkpoint, all at T_{99} in space B: two by B and one by D.

Table 6 gives paired time ratios. A ratio is formed only when both methods attain the target from that start; no finite ratio is derived from a censored cell. Wherever both reach a target from the same start, gradient-based SLSQP gets there sooner than COBYQA at the same training resolution. At L1 (B/A), the geometric-mean ratio is 3.10–3.25 in space A and 5.06–7.79 in space B. At L0 (D/E), it is 2.45–2.75 and 3.14–7.20. All 45 finite optimizer-contrast pairs (by space, target and start) favor SLSQP, with ratios from 1.46 to 21.5. The coarse mesh shortens the time by a smaller factor: the geometric-mean E/A ratio is 0.62–0.79 and D/B is 0.45–0.56. Against D, the inexpensive derivative-free physical competitor, A is faster by a geometric mean of 1.68–1.83 in space A and 2.25–4.32 in space B. This contrast changes both factors; one of its 25 pairs is a near tie (0.99,

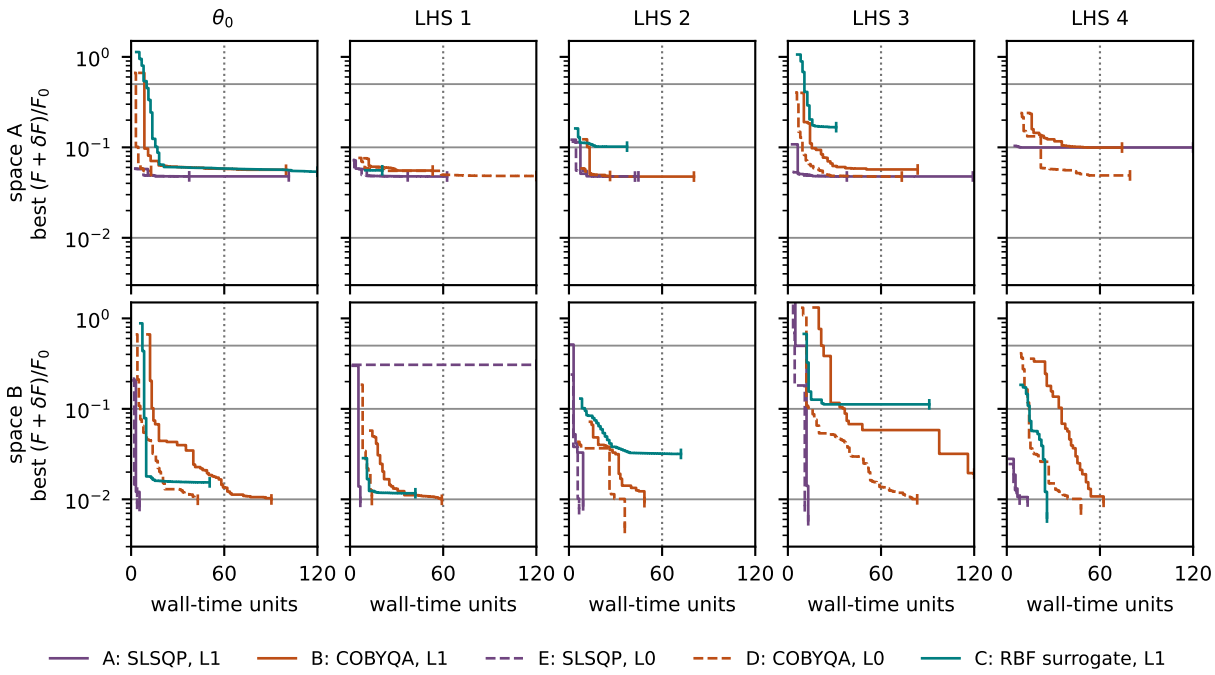


Figure 8: Best credited $(F_{\text{val}} + \delta F)/F_0$ versus charged wall time, repeat 1, at common L2/order-93/20-angle validation. Grey lines mark T_{50}, T_{90} and T_{99}; the dotted line marks the 60-unit checkpoint. Each curve ends at the run's actual stop (tick). A missing curve means the run had no credited validation.

space-A LHS 2 at T_{90}: D is 0.75% faster, within the timing noise between repeats). We conjecture that part of the larger space-B ratios comes from COBYQA's initial interpolation set, which costs $2n + 1 = 37$ training evaluations before its first model step there, against 17 in space A.

The speed of E has a reliability cost. E never produces a credited validation from space-A LHS 4. From space-B LHS 1 it reaches T_{50} but then stalls at $0.28F_0$, missing T_{90} and T_{99}. In four start–target cells, D reaches a target that E misses. A, by contrast, reaches every target that any method reaches, from every start. C attains T_{99} only once in space B and reaches T_{90} from two of five space-A starts.

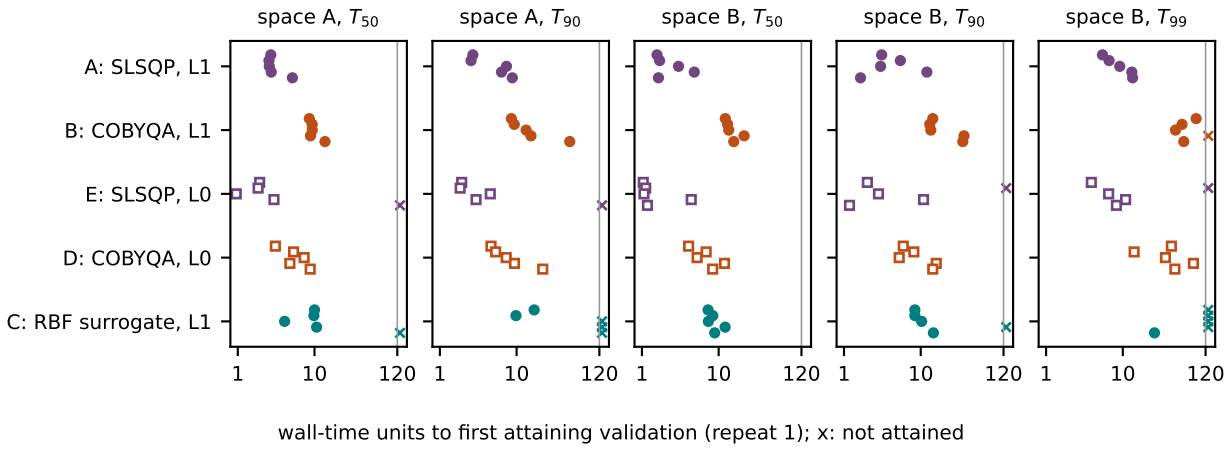


Figure 9: Charged wall time to the first attaining validation, repeat 1, one marker per start, for every target attained by some run. Filled markers: L1 training; open markers: L0 training; crosses: not attained within 120 units.

Table 5: Charged wall-time units to the first attaining validation, repeat 1. A dash means the target was not attained within 120 units. T_{99} was attained by no run in space A.

Space, target	Method	θ_0	LHS 1	LHS 2	LHS 3	LHS 4
A, T_{50}	A	2.7	2.5	2.6	2.7	5.1
	B	8.5	9.3	9.3	8.9	13.7
	E	1.9	1.8	1.0	3.0	—
	D	3.1	5.3	7.3	4.8	8.8
	C	10.0	9.8	4.1	10.6	—
A, T_{90}	A	2.7	2.5	7.4	6.4	8.8
	B	8.5	9.3	13.3	15.4	49.3
	E	1.9	1.8	4.5	3.0	—
	D	4.6	5.3	7.3	9.4	21.9
	C	16.9	9.8	—	—	—
B, T_{50}	A	1.6	1.7	3.0	4.8	1.6
	B	12.2	13.1	13.5	21.5	15.7
	E	1.0	1.1	1.1	4.4	1.2
	D	4.1	6.9	5.2	11.9	8.4
	C	7.3	8.4	7.3	12.1	8.9
B, T_{90}	A	3.1	5.4	3.0	12.0	1.6
	B	14.4	13.1	13.5	36.6	35.3
	E	2.0	—	2.8	10.9	1.2
	D	5.9	8.1	5.2	15.9	14.3
	C	8.4	8.4	10.3	—	14.6
B, T_{99}	A	5.4	6.6	9.1	13.1	13.4
	B	90.4	59.1	48.6	—	62.3
	E	3.9	—	6.5	10.9	8.3
	D	43.0	14.1	36.0	83.3	47.8
	C	—	—	—	—	25.9

Table 6: Paired per-start time ratios (repeat 1). B/A and D/E vary the optimizer at fixed resolution; E/A and D/B vary the resolution; D/A compares A with the coarse derivative-free competitor. Ratios above one favor the denominator. Geometric means use the finite pairs only, so the D/E and E/A means omit the starts where E failed to attain the target.

Space, target	Ratio	Finite pairs	Geometric mean	Per-start ratios
A, T_{50}	B/A	5	3.25	3.18, 3.66, 3.63, 3.25, 2.66
	D/E	4	2.75	1.60, 2.90, 7.68, 1.61, —
	E/A	4	0.68	0.72, 0.72, 0.37, 1.09, —
	D/B	5	0.56	0.36, 0.57, 0.78, 0.54, 0.64
	D/A	5	1.83	1.15, 2.09, 2.85, 1.75, 1.71
A, T_{90}	B/A	5	3.10	3.18, 3.66, 1.80, 2.43, 5.61
	D/E	4	2.45	2.41, 2.90, 1.61, 3.17, —
	E/A	4	0.62	0.72, 0.72, 0.61, 0.47, —
	D/B	5	0.54	0.54, 0.57, 0.55, 0.61, 0.44
	D/A	5	1.68	1.72, 2.09, 0.99, 1.48, 2.49
B, T_{50}	B/A	5	6.49	7.76, 7.70, 4.51, 4.46, 9.59
	D/E	5	4.69	3.90, 6.14, 4.93, 2.72, 7.09
	E/A	5	0.63	0.66, 0.66, 0.35, 0.91, 0.72
	D/B	5	0.46	0.33, 0.52, 0.39, 0.56, 0.53
	D/A	5	2.97	2.58, 4.04, 1.75, 2.48, 5.09
B, T_{90}	B/A	5	5.06	4.64, 2.41, 4.51, 3.05, 21.54
	D/E	4	3.14	2.94, —, 1.86, 1.46, 12.17
	E/A	4	0.79	0.65, —, 0.94, 0.91, 0.72
	D/B	5	0.45	0.41, 0.62, 0.39, 0.44, 0.41
	D/A	5	2.25	1.91, 1.50, 1.75, 1.33, 8.74
B, T_{99}	B/A	4	7.79	16.65, 8.92, 5.33, —, 4.65
	D/E	4	7.20	11.05, —, 5.50, 7.65, 5.79
	E/A	4	0.72	0.72, —, 0.72, 0.83, 0.61
	D/B	4	0.50	0.48, 0.24, 0.74, —, 0.77
	D/A	5	4.32	7.91, 2.12, 3.94, 6.36, 3.56

Final refinement. The preregistered selection takes, for each method and space, the lowest-F_{val} feasible design over both repeats, and the earliest repeat-1 crossing of each attained target. With θ_0 in each space and identical designs merged, this gives 33 designs. Each received the six registered checks: the primary L2 validation, L3, a 40-angle grid, Newton tolerance 10^{-12}, an independently solved grid shifted by 0.5°, and held-out currents 0.25 and 0.75. The 15 designs with zero stator offsets also received the conforming outer-band Arkkio check on fresh meshes. All 213 checks completed (Table 7). These checks qualify the selected designs only; they do not certify each of the study's 111 repeat-1 target crossings.

The registered pipeline begins by capturing the closed study on its original host. After that host had been restarted, the capture tool repeatedly refused on host-boot attestation details unrelated to the data. The refinement instead used a copy of the study outputs whose integrity is established by a SHA-256 inventory of every output file, taken when the session finished, before the host was restarted, and independently timestamped. It ran on a different host (Apple M3 Ultra, macOS arm64). For the 16 selected designs validated on the moved reference mesh, the primary check reproduces the study's F_{val} to below 2.2×10^{-12} relative. The other 15 candidates needed a fresh mesh, which Gmsh triangulates differently on the two platforms; their F differs by at most 1.7×10^{-3} relative. The remaining two designs are θ_0 in each space, which reproduces F_0. The fresh-mesh

differences measure the sensitivity of F to remeshing and are far below the 10% allowance δF.

Spatial refinement is the largest change. Over all 33 designs, L3 moves F by -4.0% to $+6.3\%$. Over the 25 target crossings alone, the range is -4.0% to $+1.1\%$. The largest increase belongs to the overall best design, D's in space B, whose F rises from $0.0044F_0$ to $0.0047F_0$. Angular and nonlinear effects are small: the 40-angle and shifted grids change F by at most 0.11%, and the tighter Newton tolerance by less than 1.1×10^{-5} relative. The conforming estimator gives 1.1% to 4.4% more ripple than the order-93 coupling on the space-A designs, and 2.0–2.1% more at θ_0. All 25 target crossings still satisfy their target, allowance included, at L3; the smallest remaining margin is 1.2% of the target. Every candidate also satisfies the full-current mean floor in the primary, L3, 40-angle, tight and shifted checks, the closest by 0.012%. The refinement therefore confirms the selected crossings and leaves the time-to-target comparison unchanged.

The held-out currents test the ripple reduction outside the three training currents. We compare each design's objective at currents 0.25 and 0.75, relative to θ_0's at the same currents, with its relative objective at the training currents. Across the candidates, the held-out ratio is 0.65 to 1.73 times the training-current ratio; values above one mean a smaller reduction at the held-out currents. For the five best space-A designs it is 1.07–1.09. In space B, four of the five best designs keep less of their reduction: their remaining objective fraction at the held-out currents is 1.23–1.73 times that at the training currents, while for B's best design it is 0.89. D's best design has $0.0044F_0$ at the training currents but 0.0076 of θ_0's value at the held-out currents. Part of the extra reduction from space B's added freedom is therefore specific to the three training currents.

Table 7: Final checks of the selected designs. "best" marks a method's representative; "A T_{90}" marks the earliest repeat-1 crossing of that target by that method. The L2 column is the primary check on the refinement host. L3 and conforming changes are in percent; the 40-angle, tight and shifted changes are relative, in units of 10^{-4}. "held-out" is the objective at currents 0.25 and 0.75 relative to θ_0's at the same currents. A dash means the conforming check does not apply because stator offsets are nonzero.

Design	F_{L2}/F_0	L3 (%)	g40 (10^{-4})	tight (10^{-4})	shifted (10^{-4})	conf. (%)	held-out
Space A							
θ_0	1.0000	−1.3	0.0	0.0	+0.1	+2.1	1.0000
best A	0.0432	−0.4	0.0	0.0	+0.1	+4.4	0.0472
best B	0.0432	−0.5	0.0	0.0	0.0	+4.4	0.0472
best C	0.0489	−0.7	+0.2	0.0	+0.5	+4.1	0.0524
best D	0.0432	−0.5	0.0	0.0	0.0	+4.4	0.0472
best E	0.0434	−0.6	+0.4	0.0	+0.9	+4.4	0.0473
A T_{50}, A T_{90}	0.0653	−2.9	+0.7	0.0	+1.5	+1.1	0.0686
B T_{50}, B T_{90}	0.0881	−2.3	+1.3	0.0	+2.6	+3.2	0.0866
C T_{50}	0.1472	−4.0	+0.6	0.0	+1.3	+3.3	0.1508
C T_{90}	0.0507	−0.9	+0.2	0.0	+0.5	+4.0	0.0543
D T_{50}	0.0920	−2.2	+1.5	0.0	+3.0	+3.3	0.0899
D T_{90}	0.0594	−2.3	+1.3	0.0	+2.6	+1.6	0.0609
E T_{50}	0.1105	−2.0	+0.6	0.0	+1.1	+1.2	0.1104
E T_{90}	0.0660	−2.9	+0.7	0.0	+1.5	+1.1	0.0694
Space B							
θ_0	1.0000	−1.3	0.0	0.0	+0.1	+2.0	1.0000
best A	0.0053	+2.6	+0.4	+0.1	+0.5	—	0.0066
best B, B T_{99}	0.0085	−3.0	−0.4	0.0	−1.2	—	0.0076
best C, C T_{99}	0.0057	−3.7	+2.5	0.0	+5.0	—	0.0074
best D	0.0044	+6.3	+2.6	+0.1	+4.9	—	0.0076
best E	0.0071	+2.6	−1.5	0.0	−3.1	—	0.0087
A T_{50}	0.1874	+0.9	−0.3	0.0	−0.6	—	0.1538
A T_{90}	0.0255	+1.1	−0.1	0.0	−0.2	—	0.0248
A T_{99}	0.0077	+0.4	−0.1	0.0	−0.3	—	0.0096
B T_{50}	0.1854	−1.5	−0.1	0.0	−0.3	—	0.1907
B T_{90}	0.0523	−0.4	−0.2	0.0	−0.4	—	0.0505
C T_{50}	0.3958	−1.4	−0.1	0.0	−0.1	—	0.3692
C T_{90}	0.0259	+0.7	0.0	0.0	0.0	—	0.0247
D T_{50}	0.1925	−1.5	−0.1	0.0	−0.3	—	0.1971
D T_{90}	0.0393	−0.7	+0.6	0.0	+1.2	—	0.0392
D T_{99}	0.0089	+0.4	−0.5	0.0	−1.0	—	0.0137
E T_{50}	0.1961	+0.8	0.0	0.0	0.0	—	0.1567
E T_{90}	0.0223	−2.8	−5.3	0.0	−10.6	—	0.0144
E T_{99}	0.0081	+0.7	+0.2	0.0	+0.4	—	0.0112

8 Discussion and limitations

The controlled interface change explains most of the loaded ripple gap while preserving mean-torque agreement. The order ladder shows why this matters for optimization: interface truncation changes the objective even as spatial error decreases. The paired gradient comparison also measures changes in ripple sensitivities despite small discrete derivative errors.

The model is two-dimensional and magnetostatic. It does not evaluate thermal behavior, demagnetization, fatigue, end effects, losses or a complete efficiency map. The comparison uses published computational artifacts, with no new measurements on a physical motor. The forward comparisons cover the initial and saved optimized designs; local derivative tests at an interior scalar design do not validate an entire optimization trajectory. The reference refinement is at the initial design. Denser and shifted angle grids and additional currents are checked for Opt120 and, in Section 7, for the selected optimization designs; the reference refinement itself was not repeated at those designs.

The optimization study answers the question left open by the derivative checks, within its

scope. Across five paired starts, two design spaces and two training resolutions, wherever both reach a common validated target from the same start, the adjoint optimizer gets there sooner than COBYQA at matched resolution, and the coarse derivative-free competitor D does not close that gap. The ratios are not population estimates for arbitrary starts, and the optimizer contrast also includes differences in search strategy beyond gradient access. Population methods such as CMA-ES, multi-start strategies and other surrogates remain untested. No reduced physical model, such as a magnetic equivalent circuit or a waveform model, was compared: D is a coarse finite-element model, not such a baseline. In space A, every method's best design sits on the active mean-torque floor, and four of the five reach the same $0.043F_0$ (C reaches $0.049F_0$). Within this budget and these methods, the parameterization and constraint, more than the choice among A, B, D and E, set the attainable ripple there. Space B's lower ripple combines ten extra offsets with a different mesh-motion map, and part of it does not carry over to the held-out currents.

A cheap waveform or reduced physical model is an essential baseline. The linear slotless screening control in Appendix A admits a three-coefficient angular fit at one geometry. That does not establish generalization across shapes, but it cautions against inferring an optimization advantage from parameter count or adjoint cost alone.

9 Conclusion

On identical meshes of a published nonlinear motor, order-45 truncation lowers loaded torque standard deviation by 7–9% and raises cogging standard deviation by 59% relative to conforming coupling at L3. Refining the independent reference reduces the initial cogging discrepancy from 12.9% to 0.10% at matched order, although neither spatial-refinement sequence is fully converged and the remaining error is not bounded. For the saved optimized design, raising the interface order from 45 to 93 multiplies the squared-ripple objective by 3.6 while preserving a reduction above 2,400-fold. Its shorter stack and changed radial scale must be separated from current policy when comparing mean torque with the initial motor; removing both explicit size factors puts the two fixed-current means within -1.0% to $+3.7\%$ of the initial mean. Most of their saved-size deficit therefore comes from the size factor, although the normalized fixed-physical-current value remains lower. These are algebraic normalizations, not reoptimized designs. Design gradients pass finite-difference and Taylor checks at two parameter points, while the cross-model ripple-sensitivity gap persists under refinement. The excitation rule reverses three mean sensitivities at the initial L1, order-45, full-current state. Derivative verification therefore does not settle objective resolution or feasibility. In the matched study, wherever both reached a target from the same start, adjoint SLSQP got there sooner than COBYQA at the same training resolution, in all 45 paired comparisons, while a coarser training mesh gave a smaller and less reliable saving. Every selected target crossing survived refinement, which changed the objective of those designs by -4.0% to $+1.1\%$. Over five starts in two fixed-size design spaces, verified adjoints shortened the time to a validated target; whether they do so for other motor problems remains to be tested.

Code and data availability

The original forward and gradient evidence is fixed at GEODE commit `910593ba`. The repository contains the raw motor runs and analysis, geometry and mesh-generation scripts, design-gradient sweeps and timing records, and solver verification logs. Candidate-validation records are pinned at `224332c5`, and independent reference reruns at `8acd21ec`. Reference data are extracted from the authors' ICEM snapshot at `b145aaf4`, a later repository revision rather than a claim to the original conference release. Source hashes, extraction checks and the upstream GPL-3.0-or-later license are retained. The paper package includes a manifest, copied figure inputs and a regeneration

script; it does not require rerunning FEM to rebuild the figures. Recomputing the physical study requires the separately documented mesh and solver commands; the benchmark READMEs record the GEODE-side Gmsh and Rust toolchain versions. The matched optimization study (Section 7) follows the optimization harness at `9486a9da`, the source of the study's evaluator, and the source of the final-refinement program at `3e4ee0c6`. The paper package contains the per-run validation histories, the 213 final-check results and the study's SHA-256 inventory of all 5,453 output files, taken when the session finished, before its host was restarted. The inventory is independently timestamped (RFC 3161 and OpenTimestamps), and each extracted run and session record is bound to it by hash. The session, calibration and anchor-recomputation records are included.

A Linear slotless screening control

The stored slotless model has 24 distinct angles. Fitting $T(\phi) = c_0 + c_1 \cos\phi + c_2 \sin\phi$, with electrical angle $\phi = p\theta_r$, mechanical rotor angle θ_r and pole-pair count p, on alternate angles and evaluating on the other twelve gives normalized held-out RMSE 3.08×10^{-10} and maximum error 6.82×10^{-10}, using the recorded analytic-amplitude normalization. This is a locked-rotor angle sweep, not synchronous operating ripple. For a fixed linear field operator and sinusoidal sources, the field lies in a small basis of source responses; a quadratic torque functional can contain only the corresponding finite set of harmonics. The particular model needs just the three fitted coefficients. This is an angular interpolation test at one geometry, not a shape-gradient test or evidence of generalization across designs.

B Design points for derivative checks

Table 8 identifies the scalar derivative-study coordinates and the saved Opt120 coordinates. Values are rounded for display; the accompanying records retain full precision. The operating angle is stored in degrees and converted to radians in the current equation. These coordinates describe the scalar design space; the saved reference optimum additionally contains surface-control offsets.

Table 8: Initial and interior perturbed design points for derivative checks, alongside Opt120 for the forward validation. Lengths refer to the unscaled geometry; surface-control offsets are retained in the data.

Parameter	Meaning	Unit	Initial	Perturbed	Opt120
MAG	Pole-axis iron depth above magnet	mm	7.0000	6.5506	6.1011
MH	Radial magnet height	mm	7.0000	4.8722	2.7444
MW	Magnet width	mm	19.0000	19.8670	20.7340
SW1	Stator tooth width	mm	4.0000	4.0309	4.0619
SW2	Slot-opening width	mm	2.3000	1.6500	1.0001
SW3	Slot-opening depth	mm	1.0000	0.7502	0.5004
SW4	Back-iron depth on slot axis	mm	8.2500	6.8418	5.4336
SD1	Stator outer diameter	mm	135.0000	132.6466	130.2931
Sr1	Slot-bottom fillet radius	mm	1.0000	1.4999	1.9998
LENGTH	Axial length	mm	100.0000	90.0003	80.0005
ϕ_0	Electrical current offset	°	0.0000	-5.8231	-11.6463
ScaleR	Radial similarity scale	1	1.0000	1.0239	1.0477

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